Heat energy

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the heat energy needed.

- $\Delta T = 100 - 18 = 82$ °C
- $E_h = 4180 \times 1.75 \times 82 = 599\,830$ J
- = 600 kJ

2 A kettle holds 1.5 litres of water at 5 °C. Calculate the heat energy needed to raise its temperature to each value.

- 50 °C: $\Delta T = 45$; $4180 \times 1.5 \times 45 = 282\,150$ J = 280 kJ
- 40 °C: $\Delta T = 35$; $4180 \times 1.5 \times 35 = 219\,450$ J = 220 kJ
- 80 °C: $\Delta T = 75$; $4180 \times 1.5 \times 75 = 470\,250$ J = 470 kJ
- 100 °C: $\Delta T = 95$; $4180 \times 1.5 \times 95 = 595\,650$ J = 600 kJ

3 57 kJ of heat energy is supplied to 1.7 kg of oil with a specific heat capacity of 2.7 kJ/kg K. The oil starts at 3 °C. Calculate its final temperature.

- c = 2700 J/kg K, $E_h = 57\,000$ J
- $\Delta T = E_h \div (cm) = 57\,000 \div (2700 \times 1.7) = 12.4$ °C
- Final temperature = 3 + 12.4 = 15 °C

4 Calculate the energy needed to raise the temperature of 750 g of water from 34 °C to 91 °C.

- m = 0.75 kg, $\Delta T = 57$ °C
- $E_h = 4180 \times 0.75 \times 57 = 178\,695$ J = 180 kJ

5 An electric heater supplies 13 500 J to a 0.5 kg metal block, raising its temperature from 20 °C to 80 °C. Calculate the specific heat capacity of the metal.

- $c = E_h \div (m\Delta T) = 13\,500 \div (0.5 \times 60)$
- c = 450 J/kg K